Testing Code for Idempotency

There's no general way to test a piece of code to determine if it's idempotent. It's usually not possible to have all of your third-party service providers simulate error conditions, so testing idempotent code often involves testing that it behaves the way you intended, given the conditions you wrote it for.

The changes to `send_notification_email` were substantial. You added a new API call and complex logic to examine that call to figure out what to do. The example app doesn't use mocking but since it does have the fake mail service running, you can use it to simulate the situation you're checking for.

The core of the test will call `send_notification_email`, and will expect a previously-used `email_id` to be set on the order. The test should also expect that no email has been sent from the email service.

To test both of those assertions, you'll need to trigger an email prior to calling the test by using `EmailServiceWrapper` directly. You'll also need to have it query for sent emails so you can count the number of emails both before and after the test.

It's a bit complicated, but here's what it will look like:

snapshots/3-4/sidekiq-book/test/services/order_creator_test.rb

```
    refute_nil resulting_order.fulfillment_request_id
  end

» test "send_notification_email uses existing email" do
»   email_service_wrapper = EmailServiceWrapper.new
»
»   order = Order.create!(
»     email: "pat@example.com",
»     address: "123 Main St",
»     quantity: 1,
»     product: create(:product),
»     user: create(:user),
```